

1.3 - Wake Functions

- The ME for the fields generated by a source particle of transv. coord.'s (x_0, y_0) and velocity v along the longitudinal direction z
- + Boundary Conditions of continuity for $D^\perp, B^\perp, \vec{E}^\parallel$ and $\vec{H}^\parallel$
- + Expression for momentum change of the witness particle (determ. using the mentioned approximations)

$\Rightarrow$ Fields $\vec{E}$ and $\vec{B}$ from the passage of the source part. are uniquely determined

$$\Delta \vec{p}(\vec{p}_s, \vec{p}_w, z) = \int_{-\infty}^{\infty} dt \vec{F}(\vec{p}_s, \vec{p}_w, s, t) \Big|_{s=vt-z}$$

where: $\vec{p}_s = (x_s, y_s)$, $\vec{p}_w = (x_w, y_w)$, $\vec{F} = q \left[E_s \hat{s} + (E_x + v B_y) \hat{x} + (E_y - v B_x) \hat{y} \right]$

$\vec{E} = \vec{E}(\vec{p}_w, vt-z)$
 $\vec{B} = \vec{B}(\vec{p}_w, vt-z)$

In theory:

$\Rightarrow$ Everything we need to compute the total momentum variation of each witness particle due to the effect of all other (source) particles!

Nonetheless, we can use some additional definitions that will be very handy when one seeks to indeed perform such calculations (and explore symmetries / simplifications).

The first of them are the wake functions (or wakes):

$$(w_x, w_y, w_{||}) = \frac{v}{qQ} (\Delta p_x, \Delta p_y, -\Delta p_s) \sim [V/c] = [J/c^2]$$

$\sim m v^2 [kg m^2/s^2]$
 $\sim m \cdot a \cdot d [kg m^2/s^2]$
 $p \cdot v \sim F \cdot d$

functions independent of the charges involved!

$$q_w \sim q$$

$$q_s \sim Q$$

w_i 's depend only on the "structure" that generates the wake fields